

Arjun Hariharan

Cell: (346) 857-4166 | Email: cognitivetech52@gmail.com |

GitHub - <https://github.com/Anomaly-1> | Portfolio - <https://portfolio-of-arjun.vercel.app>

EDUCATION

Clements High School

Sugarland, Texas

Expected Graduation: 2027

GPA: 4.36 W / 3.8 U — SAT: 1540

- Ranked within the top 20% of class.
- **Relevant AP Coursework:** AP Computer Science A (5), AP Physics (4), AP Precalculus.
- Harvard CS50x, CS50P, and CS50AI; Russian School of Mathematics through calculus; PACT.

RESEARCH AND WORK EXPERIENCE

Software Engineering Intern

June 2025 – Present

Jdable (Nonprofit)

Remote

- Engineered full-stack IoT and mobile apps for Smart Cane and Autonomous Wheelchair, integrating GPS, Bluetooth, and LiDAR for real-time navigation for 1,000+ users.
- Developed software interfaces for low-cost assistive hardware, including a \$10 braille keyboard deployed to 250+ students across 3 countries.

Lead Programmer

Sept 2025 – Nov 2025

Educational Game Startup

Remote

- Led development of a beta educational platform from concept through public release, scaling backend systems to support over 3,000 users in the first week.

Instructor / Mentor

Jan 2025 – Present

Independent Academic Instruction

Remote

- Designed and taught SAT-focused mathematics curricula, adapting instruction to individual learning needs and explaining complex concepts clearly.

Independent Researcher

Nov 2023 – Present

Machine Learning and Computational Intelligence

Remote

- Researching Short-Term Synaptic Plasticity in Spiking Neural Networks for energy-efficient adaptation to non-stationary environments.
- Designing experiments to evaluate synaptic mechanisms in improving network robustness and reducing computational overhead.

SELECTED PROJECTS

Custom 7in FPV Delivery Drone — Embedded Systems, 3D Printing, Firmware

- Designed and assembled a 7-inch FPV drone with custom 3D-printed retractable claw mechanism for object manipulation and hazardous-area delivery.
- Integrated flight controller firmware, power distribution, and sensor systems; performed iterative field testing and soldering.

Unbound — iOS, Swift, Accessibility AI

- Built an iOS accessibility app supporting blind, deaf, and dyslexic users with read-aloud, real-time transcription, and text-to-speech navigation.
- Integrated OCR for dyslexic-friendly fonts and wrapped Gemini Flash API for contextual AI assistance, evaluating on-device latency tradeoffs.

Tetris AI — Python, Pygame, Heuristic Search

- Implemented a heuristic-based AI agent achieving superhuman, infinite-score performance; optimized board evaluation functions and piece-placement logic.

VOLUNTEERING EXPERIENCE

Web Development Consultant

2024 – Present

Local Small Businesses

Sugarland, Texas

- Designed and deployed responsive web platforms for local businesses lacking digital presence, increasing online visibility.
- Managed end-to-end development including UI/UX design, domain registration, and SEO optimization to drive local traffic.

Logistics and Event Volunteer

2023 – 2024

Student Council

Sugarland, Texas

- Assisted with school event preparation, coordinating logistics and operations for large-scale student body activities.

EXTRACURRICULAR ACTIVITIES

Marching Band <i>Alt to Top Band Section Leader</i> Led 10-person section and 150-member band to 1st Overall and 39/242 at State. Produced film; optimized logistics for 50+ recruits.	2023 – Present
Technological Student Association (TSA) <i>Team Lead</i> 3-year State qualifier. Led 6-person teams building AI, hardware and web apps. Mentored underclassmen by authoring 50+ technical guides.	2023 – Present
VEX V5 Robotics <i>Sole Programmer</i> - Programmed C++ sensor fusion and pathing algorithms. Led team to State, winning Design and Judges' Awards. - Mentored 10 novices to scale team proficiency and authored award-winning engineering documentation.	2024 – Present
JETS - SciOly and SciBowl <i>Competitor and Web Architect</i> 3-year SciOly top-3 medalist. Mentored build teams. Architected web platform digitizing 500+ resources and tracking historical performance.	2023 – Present
Logic Club <i>Vice President and Cofounder</i> Cofounded club. Authored discrete math and circuitry curriculum. Deployed 100+ weekly logic challenges school-wide to build analytical skills.	2025 – Present
Academic Decathlon <i>Math and Science Captain</i> 1st in Math/Sci. Mastered 10 subjects, anchoring school's 1st-ever State team. Authored study guides and led marketing to scale recruitment.	2024 – Present
Technical Passion Development <i>Independent Engineer</i> Built 10+ full-stack apps. Created AI/software systems for reliability plus hardware designs for custom robots. Self-directed engineering.	2022 – Present

HONORS AND AWARDS

Hackathon Awards 1st Place, DAIS International Hackathon Won 1st globally for an AI/algorithmic solution, excelling in technical innovation. 3rd Place, HackVortex Hackathon Best of Water, Kurius Hacks	<i>International, 9th</i> <i>2024</i> <i>2024</i>
Academic and Technical Competitions USA Computing Olympiad (USACO) Silver Solved complex C++ challenges to reach Silver, placing in top 15% of global coders. TSA State Finalist (3x) Qualified 3 years across 5+ AI, web, and engineering events, placing in the top tier. VEX Robotics State Design Award Awarded for innovative C++ algorithms and rigorous engineering log vs. 80+ teams. Academic Decathlon Math/Sci Awards Anchored school's 1st State team. Placed 1st in Math and Science across 10 subjects. Science Olympiad Top-Five Finishes (10+) Judges and Design Award (x2), VEX Robotics Marching Band State Qualifier Placed 39th out of 242 bands.	<i>National, 10th</i> <i>State, 9th to 11th</i> <i>State, 11th</i> <i>State, 11th</i> <i>State/Regional, 2023 to Present</i> <i>State/Regional, 2024 to Present</i> <i>State, 2025 to 2026</i>

TECHNICAL SKILLS

Programming Languages: Python, C++, TypeScript; familiarity with Rust, C#, Java
AI / Machine Learning: PyTorch, TensorFlow, neural networks, transformers, spiking neural networks, reinforcement learning
Full-Stack Development: React, Next.js, Vue.js, Svelte, Tailwind CSS, Django, Flask, FastAPI, Supabase, Firebase, MongoDB, SQL
Embedded Systems: Arduino, Raspberry Pi, VEX robotics, sensors, motors, GPIO, firmware development
Hardware: Soldering, circuit design, basic CAD, electrical systems integration, KiCAD